Септември 2015

$$\int \frac{(x-3)}{x(x^2+2)}dx,$$ интеграл от рационална ф-я

$$\int \frac{P(x)}{Q(x)}dx.$$

Стъпки

1) ако степента на P е поне степента на Q ($\deg P \geq \deg Q$) то делим P на Q с частно и остатък

2) Получената нова рац. ф-я разбиваме на елем. дроби.

Деление многочленов

$P = x^4 + x - 4$, $Q = x^3 + 2x^2 + 3x + 3$

$x^4 + x - 4 : x^3 + 2x^2 + 3x + 3 = \underbrace{x - 2}_{\text{частное}}$

$-\ x^4 + 2x^3 + 3x^2 + 3x$

$-2x^3 - 3x^2 - 2x - 4$

$-\ -2x^3 - 4x^2 - 6x - 6$

$\underbrace{x^2 + 4x + 2}_{\text{остаток}}$

$$\frac{x^4 + x - 4}{x^3 + 2x^2 + 3x + 3} = x - 2 + \frac{x^2 + 4x + 2}{x^3 + 2x^2 + 3x + 3}$$

Разлагане на елементарни дроби

$$\frac{(x-2)}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$$, намираме A, B, C с метода на неопр. коеф.

осв. от знаменателя

$$(x-2) = A(x^2+2) + (Bx+C)x$$

$$x-2 = (A+B)x^2 + Cx + 2A$$

$x^2: \; A+B=0$

$x^1: \; C=1$ $\Rightarrow A=-1, B=1, C=1 \Rightarrow \frac{x-2}{x(x^2+2)} =$

$x^0: \; 2A=-2$

$$= \frac{-1}{x} + \frac{x+1}{x^2+2}$$

$$\int \frac{x-2}{x(x^2+2)} = \int -\frac{1}{x} + \frac{x+1}{x^2+2}\,dx = \int -\frac{1}{x}\,dx + \int \frac{x+1}{x^2+2}\,dx$$

$$\int -\frac{1}{x}\,dx = -\int \frac{1}{x}\,dx = -\ln|x|$$

$$\int \frac{x+1}{x^2+2}\,dx = \int \frac{(x)+1}{x^2+2}\,dx = \int \frac{x}{x^2+2} + \frac{1}{x^2+2}\,dx = \int \frac{x}{x^2+2}\,dx + \int \frac{1}{x^2+2}\,dx =$$

$$= \int \frac{1}{x^2+2}\,d\left(\frac{x^2}{2}\right) + \int \frac{1}{x^2+(\sqrt{2})^2}\,dx = \frac{1}{2}\int \frac{1}{x^2+2}\,d(x^2+2) + \frac{1}{\sqrt{2}}\operatorname{arctg}\frac{x}{\sqrt{2}}$$

$$= \frac{1}{2}\ln(x^2+2) + \frac{1}{\sqrt{2}}\operatorname{arctg}\frac{x}{\sqrt{2}} + C$$

$$\frac{R(x)}{(x-2)(x+3)^3(x^2+2x+4)^2} = \frac{A}{x-2} + \frac{B_1}{x+3} + \frac{B_2}{(x+3)^2} + \frac{B_3}{(x+3)^3} +$$

$$\deg R < 8 \qquad + \frac{C_1x+D_1}{x^2+2x+4} + \frac{C_2x+D_2}{(x^2+2x+4)^2}$$

Ден 2019

$$\int_1^8 \frac{dx}{\sqrt[3]{x}+x}$$

Смяна на променливите

$$\int_a^b f(x)dx \overset{x=\varphi(t)}{=\!=\!=} \int_\alpha^\beta f(\varphi(t)) \cdot \varphi'(t)\,dt$$

където $\varphi(\alpha) = a$, $\varphi(\beta) = b$

и $\varphi : [\alpha, \beta] \to [a, b]$

$t \in (\alpha, \beta) \Rightarrow \varphi(t) \in [a, b]$

$$\int_1^8 \frac{dx}{\sqrt[3]{x}+x} \overset{x=t^3}{=} \int_1^2 \frac{1}{t+t^3}\cdot(t^3)'\,dt = \int_1^2 \frac{3t^2}{t+t^3}\,dt =$$

$$= \int_1^2 \frac{3t}{1+t^2}\,dt = 3\int_1^2 \frac{1}{1+t^2}\,d\left(\frac{t^2}{2}\right) = \frac{3}{2}\int_1^2 \frac{d(t^2+1)}{t^2+1} =$$

$$= \frac{3}{2}\ln(t^2+1)\Big|_{t=1}^{2} = \frac{3}{2}(\ln 5 - \ln 2)$$

Пр1 $\int_0^{1/2} \arcsin x\,dx = \left| x = \sin t \right| = \int_0^{\pi/6} \arcsin(\sin t)\cdot(\sin t)'\,dt =$

$= \int_0^{\pi/6} t\cdot\cos t\,dt$

Пр2. $\int \frac{dx}{\sqrt[3]{x}+\sqrt{x}} \overset{x=t^6}{=\!=} \int \frac{1}{t^2+t^3}\cdot 6t^5\,dt$

$$\int \frac{x\,dx}{2\sqrt{\frac{x-1}{x+2}} + 5\sqrt{\frac{x+2}{x-1}}} = \int \underbrace{\frac{\frac{-1-2t^{10}}{t^{10}-1}}{t^5 + \frac{1}{t^2}} \cdot \varphi'(t)}\, dt$$

рац. ф-я.

$$\frac{x-1}{x+2} = t^{10}, \quad x = \underbrace{\frac{-1-2t^{10}}{t^{10}-1}}_{\varphi(t)}$$

$\sqrt{\frac{x^2-1}{x+1}}$ | Тази схема не върши работа

Дан 2018

$$\int_0^{\frac{\pi}{2}} \frac{1}{2+\sin x}\,dx$$

Интеграли от рац. ф-ии на $\sin x$ и $\cos x$

$$\cos 2x = \frac{2\cos^2 x - 1}{2}$$

Универс. субст.

$$x = 2\operatorname{arctg} t, \quad t = \operatorname{tg}\frac{x}{2}$$

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2}{1+t^2}\,dt$$

$$\int_0^{\pi/2} \frac{1}{2+\sin x}\,dx \overset{y.c.}{=} \int_0^1 \frac{1}{2+\frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2}\,dt =$$

$$2\operatorname{arctg} 1 = \frac{\pi}{2}$$

$$= \int_0^1 \frac{1}{1+t+t^2}\,dt = \int_0^1 \frac{1}{\left(t+\frac{1}{2}\right)^2+\left(\sqrt{\frac{3}{4}}\right)^2} d\left(t+\frac{1}{2}\right) =$$

$$1+t+t^2 =$$

$$1+2\cdot t\cdot\frac{1}{2}+t^2+\frac{1}{4}-\frac{1}{4} =$$

$$= \left(t+\frac{1}{2}\right)^2+\frac{3}{4}$$

$$= \frac{1}{\sqrt{\frac{3}{4}}} \cdot \operatorname{arctg}\left(\frac{t+\frac{1}{2}}{\sqrt{\frac{3}{4}}}\right)\Bigg|_0^1 =$$

$$= \frac{2}{\sqrt{3}} \left(\operatorname{arctg} \sqrt{3} - \operatorname{arctg} \frac{1}{\sqrt{3}} \right) = \frac{2}{\sqrt{3}} \cdot \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{2\pi}{\sqrt{3} \cdot 6}$$

Пр. $\displaystyle\int_0^{2\pi} \frac{1}{2 + \sin x}\, dx = \int_0^{\pi} + \int_{\pi}^{2\pi} =$

$$= \lim_{x \to \pi^-} F(x) - F(0) + F(2\pi) - \lim_{x \to \pi^+} F(x)$$

$$\lim_{x \to \pi^-} \operatorname{arctg}\left|\tan\left(\frac{x}{2}\right)\right| = +\frac{\pi}{2}$$

$$\lim_{x \to \pi^+} \operatorname{arctg}\left|\operatorname{tg}\left(\frac{x}{2}\right)\right| = \frac{-\pi}{2}$$

$$\int \frac{1}{\cos 2x + \sin x + 2} \, dx$$

$$\cos 2x = \frac{2\cos^2 x - 1}{2}$$

$$\sin 2x = 2\sin x \cdot \cos x$$

$$\int \frac{3x+2}{x^2+2x+4}\,dx = \int \frac{3x+5}{(x+1)^2+3}\,dx =$$

$$= \int \frac{3(x+1)+2}{(x+1)^2+3}\,dx = \int \frac{3(x+1)}{(x+1)^2+3}\,d(x+1) - \int \frac{1}{(x+1)^2+3}\,dx$$

$$= \frac{3}{2}\int \frac{d\left((x+1)^2+3\right)}{(x+1)^2+3} - \int \frac{1}{(x+1)^2+(\sqrt{3})^2}\,d(x+1)$$

$$\| \qquad\qquad \| \qquad\qquad \| \qquad\qquad \|$$

$$\ln\left|(x+1)^2+3\right| \qquad \frac{1}{\sqrt{3}}\operatorname{arctg}\left(\frac{x+1}{\sqrt{3}}\right)$$

$$\int \frac{1}{(x^2+1)^2} dx$$

$$\int e^x \sin x \, dx$$

$\forall\ x, y \in (0,1],\ x \neq y$

$|\arcsin x - \arcsin y| > |x - y|$

Т. Лагранж $(\arcsin x - \arcsin y) = \arcsin'(\eta)(x-y)$

$|\arcsin x - \arcsin y| = \frac{1}{\sqrt{1-\eta^2}} \cdot |x-y|$, где η между x и y

$\Rightarrow \eta \in (0,1) \Rightarrow$

$$\frac{1}{\sqrt{1-\eta^2}} > 1 \Rightarrow$$

$$|\arcsin x - \arcsin y| = \frac{1}{\sqrt{1-\eta^2}}|x-y| > |x-y|$$

За функцията

$$|\arctg x - \arctg y| < |x-y|, \quad \forall x, y > 0, \; x \neq y$$

$$\int x^{\alpha} dx = \frac{x^{\alpha+1}}{\alpha+1} + C \quad \alpha \neq -1$$

$$\int \frac{1}{x} dx = \ln x$$

$$\int e^x dx = e^x$$

$$\int \sin x dx = -\cos x$$

$$\int \cos x dx = \sin y$$

$$\int \frac{1}{\sqrt{x^2+a}} dx = \ln\left|x + \sqrt{x^2+a}\right|$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right)$$

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \operatorname{arctg}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\cos^2 x} dx = \operatorname{tg} x + C, \ldots$$

$$\int e^{-2x}\,dx = \frac{1}{-2}\underbrace{\int e^{-2x}\,d(-2x)}_{e^{-2x}} = \frac{1}{-2}\,e^{-2x} + C$$

$$e^{-2x} = e^{-2x}\cdot(-2)$$

$$(e^{y})'_{y} = e^{y} \qquad \left(e^{g(x)}\right)' = e^{g(x)}\cdot g'(x)$$